ACTIVITY DECISION MAKING.	NAME : TANUSHRI S.V. REG NO : 192821163 CODE : ITA0802 COURSE : COMPUTER VISION

1. MEDICAL IMAGE ENHANCEMENT :

UNDERSTANDING OF PROBLEM CONTEXT :

* An x-ray image clearly shows the bones, but the darker soft tissues have poor visibility because their pixel intensities are concentrated in the lower gray level-range. The goal is to enhance the visibility of these darker regions without losing important details.

APPLICATION OF CONCEPTS :

1.) Which gray-level transformation would you recommend?

* Log Transformation.

2.) Why is it suitable?

* It enhances dark tissues by increasing the brightness of low-intensity pixels while keeping bones visible.

3. What effect will it have on the image?

* Soft tissues become clearer, contrast improves, and the X-ray is easier to analyze.

2. NIGHT - TIME SURVEILLANCE.

* A CCTV camera captures a parking lot at night. Most pixel values are concentrated between 0 and 40, making the image very dark.

1. Which transformation would improve the visibility?

* Log Transformation.

2. Why is it more suitable than negative transformation?

* Log transformation brightens dark regions and reveals hidden details, whereas negative transformation only inverts the image.

3. Predict the appearance of the enhanced image.

* The image becomes brighter, making vehicles, people, and other objects easier to see.

3. FINGERPRINT RECOGNITION.

* A fingerprint image contains many fine ridge details, but the contrast between ridges and valleys is very slow.

1. Which gray-level transformation would you apply?

   * Contrast Stretching.

2. Explain why.

   * It increases the contrast between fingerprint ridges and valleys, making fine details clearer.

3. What would happen if a log transformation were applied instead?

   * The image would become brighter, but ridge details may become less distinct, reducing recognition accuracy.